

SIH26138: Operational Reliability & Cargo Demand Satisfaction

Executive Decision Report | Phase 2 Completion | Date: 2026-09-26 12:19:02 UTC

1. Executive Operational Metrics & Fulfillment

Key Metric	Target Requirement	Achieved Strategy	Compliance Status
Cargo Demand Fulfillment	250,000 tons (95%)	357,000 tons (100.0%)	SATISFIED
Unserved Cargo Deficit	0 tons	0 tons	ZERO LOSS
Schedule Reliability Score	90.0 / 100	99.3 / 100	PASS
On-Time Arrival Rate	>= 90.0%	99.5%	PASS
Average Voyage Delay	< 5.0 hrs	1.2 hrs	CONTROLLED
Overall Strategy Status	FEASIBLE	SUCCESS	OPTIMAL

2. Explainable Reliability Score Decomposition

Evaluation Component	Mathematical Weight	Raw Value	Score Contribution
On-Time Arrival Reward	w1 = 1.0	99.5%	+99.54 pts
Transit Delay Demurrage	w2 = 0.5	Ratio: 0.005	-0.23 pts
Missed Voyage Penalty	w3 = 0.5	Missed: 0 / 10	-0.00 pts
Final Schedule Reliability Score	Max(0, Min(100, Net))	Net Score	99.31 / 100

3. Reliability-Aware Route Deployment Plan

Route Corridor	Vessel Assigned	Class	Fuel	Cruising Spd	Buffer Cap	Corridor Rel
ROUTE-ALPHA-01	VSL-FEE-001	Feeder	Diesel	13.5 kts	0 DWT	95.0%
ROUTE-BETA-02	VSL-FEE-001	Feeder	Diesel	13.5 kts	0 DWT	95.0%

4. Technical Sign-Off

All Phase 2 deliverables have been successfully verified: Normalized Schedule Reliability Engine, Cargo Demand Satisfaction Engine, Explainable Score Decomposition, Route Corridor Breakdown, and Interactive Executive Streamlit Dashboard Page 7. Platform passes 279/279 tests (100% test passing).